

Alen Reuel

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Education

National University of Singapore, Singapore

Master of Computing

- Current CGPA: 4.88/5.00

Aug 2025 – Present

Singapore, Singapore

Nanyang Technological University, Singapore

Bachelor of Engineering (Chemical and Biomolecular Engineering)

- Graduated with Honors, Highest Distinction, CGPA: 4.62/5.00
- Dean's List Award Recipient for AY 2019/20, 20/21, 21/22 (Top 5% in cohort)

Aug 2018 – May 2022

Singapore, Singapore

Work Experience

Software Engineering Intern, Backend and Machine Learning

Aleph Technologies

- Built the MLOps pipeline which automated model training, validation, deployment and monitoring using Azure ML and Prefect, that resulted in reduced training to deployment time by 80%.
- Leveraged Prefect and designed orchestration workflows to parallelize ML model training computation across remote compute on Azure. Reduced backend server load by 70% and average end-user request time by 40%.
- Identified frontend data processing bottlenecks, refactored API calls to directly upload to Azure Blob storage through signed URLs instead of data transfer via HTTP POST requests. Reduced memory-related failure events on client-side by 40-50%.

Sep 2025 – Apr 2026

Singapore, Singapore

Research Engineer, Applied Machine Learning

National University of Singapore

- Collaborated with Pfizer and led a research project on the application of machine learning for equipment calibration that reduced operational downtime in pharmaceutical plants.
- Modeled industrial processes using RNN-based models (PyTorch) that demonstrated reliable predictions with high accuracy ($< 5\%$ MAPE). Improved equipment recalibration time by 75% through faster feedback loops compared to traditional methods.
- Designed API endpoints (FastAPI) for web application for inferencing ML models and built docker images for reproducible deployments that reduced integration time for industry collaborators by 43%.
- Organized meetings twice a week to collect user feedback on UX from collaborators. Consistently resolved 90% UI concerns in React.js application prior to subsequent review cycles.

Nov 2023 – Aug 2025

Singapore, Singapore

Process and Equipment Engineer, Dry Etch

Micron Semiconductor Asia Pte Ltd., Singapore

- Conducted rigorous statistical analysis of the product line, pinpointed underlying factors contributing to KPI drifts, delivered weekly analytical reports and presented findings to cross-functional stakeholders.
- Extracted, transformed, and analyzed manufacturing datasets to optimize process and equipment performance. Identified equipment requiring preventive maintenance and ensured $> 80\%$ availability of production equipment.
- Qualified multiple pieces of equipment through structured experimentation and data analysis. Increased overall production capacity by 10%.

Aug 2022 – Feb 2023

Singapore, Singapore